11.4 Considerations

By now, it should be clear that in the context of gaining actionable insights efficiently, signal correlation plays a pivotal role. I won't pretend it is easy to achieve complete and fully automated correlation, even in the best case, where you have a multisignal backend and a single pane of glass. There are, at the time of publication, still a large number of challenges associated with implementing correlation, representing the single largest risk category in the observability space. So on one hand, you shouldn't treat them as an afterthought, and on the other hand, you should be aware of the limitations and set your expectations appropriately. For example, if you're in a platform team (internal developer platform to streamline infra and app provisioning so that one can avoid TicketOps), you may want to prioritize correlation accordingly.

> **NOTE** Here, *TicketOps* is not referring to the company but the slightly derogatory term that used to describe the process of developers creating tickets in, say, Jira, with operations roles (in the platform team) resolving them asynchronously. This may take some time (e.g., days or weeks) and is contrary to the automation mantra put forward by DevOps.

And now, without further ado, let's have a look at the concrete challenges of implementing signal correlation in your organization. I've grouped them into three buckets, and for some of the challenges, I suggest one or more remediations you could try out to mitigate the risk.

11.4.1 Early days

First off, let's review the main challenges associated with the state of observability, in general, and implementations, in particular.

LACK OF STANDARDS

APM and monitoring tools, open source and commercial alike, use different formats for the telemetry signals as well as various terminologies. This makes it challenging to correlate signals across systems.

The following is an appropriate remediation: if you want true interoperability and to avoid vendor lock-in, use OpenTelemetry.

LACK OF METADATA

To drive correlation, you need the connective tissue represented by metadata, including but not limited to EC2 metadata, Prometheus labels, Kubernetes labels, and OpenTelemetry resource and signal attributes. Few platforms and systems to date are "observability first" (designed with observability support), so it may be difficult to acquire said metadata or involve manual work.

The following are appropriate remediations:

- Evaluate (compute) platforms according to their metadata richness and degree of automation.
- Use OpenTelemetry-based systems.